LESSON 3.5

Compare Ratio Relationships

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare ratios by finding unit rates or using equivalent ratios.

WHY THIS MATTERS Two friends used the paint shown to create shades of purple paint. Whose mixture contains the most blue paint?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Unit rate *Spanish: Tasa unitaria*	A rate for just 1 of something, like cost for 1 item.	$3 per 1 pound → unit rate is $3/lb
Equivalent ratio *Spanish: Razón equivalente*	Two ratios that mean the same thing.	4:6 and 2:3 are equivalent (both simplify to 2:3)
Compare *Spanish: Comparar*	To look at ratios and see which is bigger, smaller, or equal.	3:4 vs. 2:5 — which has more of the first ingredient per unit?
Simplify *Spanish: Simplificar*	To write a ratio with smaller numbers that still compares the same way. 6:8 and 3:4 are the same ratio.	8:12 → divide both by 4 → 2:3
Common denominator *Spanish: Denominador común*	A bottom number that two fractions can share so you can compare them.	To compare 3/5 and 4/7, use LCD 35: 21/35 vs 20/35
Common multiple *Spanish: Múltiplo común*	A number that both terms can be scaled to.	2 and 3 both reach 6

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Mix A has 2 soap to 5 water. Mix B has 3 soap to 7 water. Which is stronger?

1. Common multiple of 2 and 3: 6.
2. Mix A × 3 = 6 : 15. Mix B × 2 = 6 : 14.
3. Compare water: 14 < 15.

Answer: Mix B is stronger (less water for 6 soap)

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Store A sells 2 pens for $6. Store B sells 3 pens for $12. Which is the better deal?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I scaled both ratios to _____ _____ .
- Mix _____ has _____ and Mix _____ has _____ .
- _____ is stronger because _____ .
- You cannot compare them directly because _____ .

Word bank common multiple | least common multiple | scale | match | compare | stronger | weaker | less | more | same | for every | fair

WATCH OUT!

A common mistake

A common mistake in Compare Ratios is thinking the ratio with the bigger raw number automatically wins

LESSON 3.5

Compare Ratio Relationships

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare ratios by finding unit rates or using equivalent ratios.

WHY THIS MATTERS Two friends used the paint shown to create shades of purple paint. Whose mixture contains the most blue paint?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the unit rate for each option to compare them.

Option	Ratio	Unit Rate
Juice A	6 oz for 2 servings	3 oz per serving
Juice B	10 oz for 4 servings	
Juice C	9 oz for 3 servings	

HINT Row 1 shows the move: 6 oz for 2 servings became 3 oz per serving by dividing.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

(A) Chef A
(B) Chef B
(C) They use the same amount
(D) Cannot determine

HINT You can't compare 2 cups and 3 cups directly — the chefs use different numbers of crackers.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which ratio is greater: 3:4 or 5:8?

(A) 5:8
(B) 3:4
(C) They are equal
(D) Cannot determine

HINT The second numbers (4 and 8) don't match, so the ratios aren't ready to compare yet.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

(A) Recipe B
(B) Recipe A
(C) They are the same
(D) Cannot determine

HINT Recipe B has more lemons, but it also has more water — so compare rates, not raw counts.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which is the better deal: 4 apples for $3 or 6 apples for $5?

(A) 6 apples for $5
(B) 4 apples for $3
(C) They cost the same
(D) Cannot determine

HINT 'Better deal' means the LOWER cost for one apple — not simply more apples.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read the problem** Chef Priya makes hot chocolate with 2 tbsp cocoa for 4 oz milk. Chef Owen makes hot chocolate with 3 tbsp cocoa for 8 oz milk. Whose hot chocolate is more chocolatey?
2. **Compare the cocoa** Chef Owen has 3 tbsp cocoa. Chef Priya has 2 tbsp cocoa.
3. **Draw a conclusion** Since 3 is greater than 2, Chef Owen's hot chocolate must be more chocolatey.
4. **Conclusion** Chef Owen's hot chocolate is more chocolatey.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at Step 3. The chef compared only the cocoa amounts (2 and 3) without checking whether the milk amounts were equal.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The two recipes use different amounts, so you cannot compare them directly. How did building equivalent ratios let you compare them fairly?

SENTENCE STARTER I scaled both ratios to the same amount of ___, then compared the ___; Chef ___'s recipe is more chocolatey because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.5

Compare Ratio Relationships

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare ratios by finding unit rates or using equivalent ratios.

WHY THIS MATTERS Two friends used the paint shown to create shades of purple paint. Whose mixture contains the most blue paint?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Match each scenario to the correct answer: Which ratio is greater?

Groups: Correct Comparison | Incorrect Comparison

_____ 5:6 vs. 7:9 → 5:6 is greater

_____ 2:3 vs. 4:6 → they are equal

_____ 1:4 vs. 2:5 → 2:5 is greater

_____ 3:7 vs. 2:5 → 2:5 is greater

_____ 4:5 vs. 3:4 → 3:4 is greater

_____ 1:2 vs. 3:4 → they are equal

2 **Complete the table. Show how you found each value.**

Two frosting recipes both mix drops of food coloring to make green frosting for a cake. Recipe A: 3 drops blue to 5 drops yellow. Recipe B: 4 drops blue to 7 drops yellow. Scale both to a common amount of yellow to compare which recipe is more blue.

Recipe	Blue Drops	Yellow Drops
Recipe A	3	5
Recipe B	4	7
Recipe A (scaled to 35)		35
Recipe B (scaled to 35)		35

SHOW YOUR WORK

3 **Mark and label each value on the number line.**

Place each unit rate on the number line to compare: A = $0.60/oz, B = $0.75/oz, C = $0.50/oz



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Store A sells 3 mangoes for $6. Store B sells 5 mangoes for $9. Which store offers a better price per mango?

(A) Store A
(B) Same price
(C) Store B
(D) Cannot determine

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read the problem** Compare Chef Reyes's dressing (4 tbsp oil : 6 tbsp vinegar) to Chef Tran's dressing (5 tbsp oil : 8 tbsp vinegar). Which dressing is oilier?
2. **Find a common denominator** LCD of 6 and 8 is 24.
3. **Scale each ratio** Chef Reyes: 6 × 4 = 24, so oil = 4 × 4 = 16, giving 16:24. Chef Tran: to reach 24 he also multiplies by 4: oil = 5 × 4 = 20, giving 20:24.
4. **Conclusion** Since 20 > 16, Chef Tran's dressing is oilier.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The two recipes use different amounts, so you cannot compare them directly. How did building equivalent ratios let you compare them fairly?

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LESSON 3.5

Compare Ratio Relationships

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

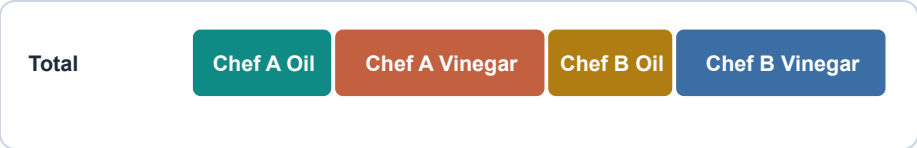
LEARNING TARGET I can compare ratios by finding unit rates or using equivalent ratios.

WHY THIS MATTERS Two friends used the paint shown to create shades of purple paint. Whose mixture contains the most blue paint?

PART 1 Understand the idea

models and meaning

1 **Use the model to solve. Show your work.**



Chef A: 2:3 → with 15 tbsp vinegar, oil = 10 tbsp. Chef B: 3:5 → with 15 tbsp vinegar, oil = 9 tbsp. Chef A uses more oil.

SHOW YOUR WORK

Answer: _____

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Kai's Mistake

1. **Read the problem** Compare 3:5 and 4:7. Which ratio is greater?
2. **Find common denominator** LCD of 5 and 7 is 35
3. **Scale the ratios** 3:5 = 21:35 (3 × 7 = 21). For 4:7, Kai multiplies by 7 again: 4 × 7 = 28, giving 28:35.
4. **Conclusion** Since 28 > 21, the ratio 4:7 is greater.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Store A sells 5 notebooks for $12. Store B sells 8 notebooks for $20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

The two recipes use different amounts, so you cannot compare them directly. How did building equivalent ratios let you compare them fairly?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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